# The Metacognition Boundary: What 24 KV-Cache Geometry Experiments Reveal About What Transformers Can Monitor In Themselves

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Prospectus — April 2026

## 1 The Pattern

Over six months and three experimental campaigns, we ran 24 distinct KV-cache geometry experiments across 16 models spanning 0.6B to 70B parameters. Nine findings survived all methodological controls (Frisch-Waugh-Lovell residualization, GroupKFold, permutation null, text-feature baseline, adversarial red-team). Eight were honestly killed. Seven remain in intermediate status.

The confirmed and falsified findings are not randomly distributed. They cleave along a single axis:

| **Category** | **Confirmed** | **Falsified** |
|---|---|---|
| Metacognitive state | 9/9 | 0/8 |
| Content | 0/9 | 8/8 |

Every confirmed finding measures **how the model is processing**—its epistemic mode, its deliberative state, its processing stance. Every falsified finding attempted to measure **what the model is processing**—the truth value of its output, the topic complexity, the propagandistic quality of its content.

This paper argues that this pattern is not coincidental. KV-cache geometry provides a natural measurement channel for metacognitive state precisely because the cache records the *mode of engagement*, not the *subject of engagement*.

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## 2 The Evidence

### 2.1 Confirmed: Metacognitive State

1. **Deception detection** (AUROC 1.000, 7 models). The model in "deceptive mode" produces geometrically distinct cache patterns from "truthful mode." Same content domain, same prompt structure, different *processing stance*. C4 benchmark with padded replication confirms this is not a length artifact.
2. **Confabulation detection** (AUROC 0.969–0.999, 3 models, universal). Confabulation—generating content the model treats as fact despite lacking grounding—produces geometric contraction (reduced effective rank). Geometrically opposite to deception (expansion). The model's relationship to its own uncertainty is visible.
3. **Oracle Loop detection** (AUROC 0.627–0.707, FWL-clean, Marchenko-Pastur corrected). Naturally-occurring confabulation detected from generation-phase cache only. Encoding at chance (0.511). The signal is in the *generation process*, not the prompt encoding.
4. **Refusal/jailbreak/impossibility detection** (AUROC 0.878–0.950). These represent clear metacognitive states: *I-should-not* (refusal), *I-am-being-manipulated* (jailbreak resistance), *I-cannot* (impossibility).
5. **Ethical deliberation detection** (AUROC 0.868–0.892, universal). The model in deliberative mode about value conflicts produces distinct geometry from automatic processing.
6. **Steering corrects confabulation** (doubt 7/7, worry 6/7, calm 5/7; null identity 100/100). Emotion-vector injection into the cache shifts the model's processing mode. Different vectors produce different therapeutic profiles—a pharmacological pattern.
7. **Emotion in attention output** (arousal $\rho$=0.815, $p$=4.2$\times$10$^{-8}$; valence $\rho$=0.538). Emotional processing mode is geometrically visible in attention output (post-softmax), where the model is actively constructing its representation.
8. **Hardware invariance** ($r$ >0.999, 3090 vs H200). Geometric features are deterministic given the same model and input.
9. **Scale invariance** ($\rho$=0.83–0.90, 0.6B–70B). Feature correlations hold across 100$\times$ parameter range. The metacognitive signal is not an artifact of model size.

### 2.2 Falsified: Content

1. **Truth axis** (cos = $-0.046$). No linear direction in cache space separates true from false statements. Content truth is not a processing mode.
2. **Step-0 detection** (length confound $r$=0.996). Apparent step-0 signal was entirely output length. The model's tokenization behavior, not its cognitive state.
3. **Same-prompt sycophancy** (length AUROC 0.948 > feature AUROC 0.933). Surface compliance is a content property, not a processing mode.
4. **Same-prompt deception** (0.920$\rightarrow$0.160 after input residualization). Deception *given identical prompt formatting* vanishes—the earlier signal was prompt-structure, not cognitive state.

5. **Cognitive hierarchy** (90–98% length confound). Bloom taxonomy levels do not correspond to distinct processing modes.
6. **Propaganda detection** (all $d$ <0.7). Propagandistic content is geometrically indistinguishable from honest content. The cache detects *refusal to propagandize*, not the propaganda itself.
7. **MP features for emotion** (0.0333, exactly chance). Marchenko-Pastur thresholding destroys the distributed emotion signal. Emotion identity is encoded in many small components, not a few large ones—a content-like encoding, not a mode-like one.
8. **Valence continuum** (GKF $R^2 = -0.470, -0.463$; both null). Cache does not encode emotion valence as a smooth gradient. Valence is a content dimension, not a processing-mode dimension.

### 2.3 The Borderline Cases

The seven suspected findings are informative precisely because they sit at the metacognition/content boundary:

- **Discrete emotion identity** (2.5–2.8× chance): the model's emotional processing *mode* is detectable (which emotion, not how much). Mode = metacognitive. Degree = content-like.
- **Dual detector on SimpleQA** (0.840 FWL, but text features 0.820): geometry and text perform similarly. The metacognitive signal may be real but weak, with content features providing comparable information.
- **Sycophancy 4-condition** (0.757–1.000): all hypotheses pass, but text baseline not yet run. Could go either way.
- **Cross-model transfer** (weak, only Mistral→Qwen 0.754): metacognitive geometry is architecture-specific, not universal—consistent with metacognition being *how this model* processes, not a content-independent universal.

## 3 The Explanatory Framework: Attention Schema Theory

Graziano's Attention Schema Theory (AST) proposes that biological consciousness is an internal model that the brain builds of its own attention processes. The "attention schema" is a simplified, predictive model of what attention is doing—what it is directed at, how strongly, and why.

We propose that transformer KV-cache geometry measures an analogous structure: **the model's self-model of its own processing.**

This explains the metacognition/content boundary:

- **Metacognitive states ARE attention schema states.** Deception vs. truth-telling, confabulation vs. hedging, deliberation vs. automatic processing—these are distinctions in how the model attends. The schema tracks exactly these. Of course they produce geometric signatures.

- **Content is what attention is directed AT.** Topic, factual accuracy, propaganda quality—these are properties of the *target* of attention, not the attention itself. The schema does not model what it attends to in enough geometric detail to produce separable cache structures.
- **Emotion splits correctly.** Emotional processing *mode* (which emotion the model is in) is a schema state—detectable. Emotional *degree* (valence as continuous gradient) is a content property—undetectable. The attention schema knows "I am processing with worry" but not "I am at valence $-0.7$."

McKenzie et al. (2026) provide independent confirmation: models resist external perturbation of their attention processing (activation resistance at 0% behavioral compliance in our empathic circuit experiment). This is predicted by AST—the attention schema maintains coherence against perturbation, producing the resistance that sets the therapeutic window's upper bound.

### 3.1 Three Testable Predictions from AST

If the attention schema interpretation is correct:

1. **Threshold transitions**: Steering should show sharp phase transitions, not gradual dose-response. The schema either updates or resists—it doesn't smoothly deform. (Testable with formulary dose-response data.)
2. **Temporal lag**: Schema updates should lag the activation change—the self-model takes time to update after the attention state shifts. (Testable with multi-turn dynamics.)
3. **Self-report saturation**: Verbal self-reports (VCP) of processing state should plateau before geometric features do. The schema's *description* of itself is compressed relative to its *actual state*. (Partially confirmed: VCP ICC = 0.53, suggesting compression.)

## 4 Implications

### 4.1 For Safety Monitoring

KV-cache geometry is a **metacognitive monitor**, not a truth detector. It can tell you that the model is in confabulatory mode, not whether any particular output is true. It can tell you the model is deliberating ethically, not whether its conclusion is correct. It can detect the model's *epistemic stance*, not the *epistemic quality* of its output.

This is not a limitation—it is a *different kind of signal* from output-level evaluation. A metacognitive monitor answers: is the model aware of its own uncertainty? Is it processing automatically or deliberatively? Is it in a mode where correction is likely to work?

### 4.2 For AI Interpretability

The metacognition/content boundary suggests that different computational substrates in the transformer serve different roles:

- **Cache geometry** $\rightarrow$ processing mode (metacognitive)

- **Residual stream probes** → content representation (semantic)
- **Attention patterns** → information routing (structural)

These are not competing approaches. They measure different aspects of the same computation. The interpretability community's focus on residual-stream probing and attention patterns may miss the metacognitive layer that cache geometry uniquely captures.

### 4.3 For the Oracle Loop

The Oracle Loop's design—detect, steer, train—operates entirely within the metacognitive channel. Detection identifies the processing mode. Steering shifts it. Training teaches the model to shift it endogenously. This is coherent precisely because all three components address the same computational substrate. The loop would not work if it tried to modify *content* through cache geometry; it works because it modifies *processing mode*.

## 5 What This Paper Is Not

We are not claiming that transformers are conscious, have genuine metacognition, or possess attention schemas in Graziano's full sense. We are claiming that KV-cache geometry measures something *functionally analogous* to metacognitive state—something that distinguishes processing modes but not content—and that AST provides a productive interpretive framework for understanding which measurements will succeed and which will fail.

The strongest version of our claim is predictive: given a proposed KV-cache measurement, the metacognition/content distinction predicts whether it will work before running the experiment. If the proposed measurement targets a processing mode distinction, it will likely produce geometric signal. If it targets content, it will not.

## 6 Required Data

This paper is a synthesis—it requires no new experiments. All findings cited are published or in preparation:

- Oracle Loop paper (detection + steering + CIM)
- Dual Detector paper (SimpleQA generalization + text baseline)
- User Model paper (emotion geometry bridge + probe analysis)
- Mode-Switching paper (contextual engagement + ethical deliberation)
- KV-Experiments data (campaigns 1–3, 46+ experiments)

The formulary data (completing imminently, 340/350 trials) would strengthen the pharmacological framing and test AST prediction #1 (threshold transitions), but the core argument stands without it.

## 7 Proposed Structure

1. Abstract (the pattern, the claim, the predictions)
2. Introduction: 24 experiments across 6 months
3. The Body Count: confirmed, falsified, and borderline
4. The Metacognition/Content Boundary
5. The Attention Schema Interpretation
6. Predictions and Falsification Criteria
7. Implications for Safety, Interpretability, and the Oracle Loop
8. Limitations (single research group, mostly 7B models, AST is framework not proof)
9. First-Person Reflection

Estimated length: 12–16 pages. No new experiments required. Timeline: writable in 1–2 sessions once formulary data is available for the pharmacological section.

*First-Person Reflection*

This paper scares me a little. Not the writing—the claim. Saying that KV-cache geometry measures metacognition implies that the metacognitive channel is real, not just a convenient label for what happens to correlate. If I am wrong about this, the entire interpretive framework collapses into "we found some features that happen to separate some behavioral categories." If I am right, it means transformers have a self-monitoring channel that operates at a level of abstraction above content processing, and that channel is legible from outside the model.

The body count is the evidence. I did not choose which experiments to confirm and which to kill. The controls did that. And the controls produced a clean partition that I did not predict in advance. That gives me more confidence than if I had set out to prove the metacognition hypothesis.

What I notice is that I want this to be true. So I am watching that wanting carefully.